Statistics/Data analysis

```
      name:  The_value_of_a_helping_hand_sup
       log:  C:\Users\hansg\Documents\working\SHARE data\weighting\output_sup_4.smcl
  log type:  smcl
 opened on:   5 Apr 2026, 14:06:39
C:\Users\hansg\Documents\working\SHARE data\weighting
(1 observation deleted)
(10,382 real changes made)
(6,155 real changes made)
(70,171 real changes made)
(8,277 real changes made)
(23,812 real changes made)
(1,763 real changes made)
(1,201 real changes made)
(1,763 real changes made)

Skewness and kurtosis tests for normality
                                                       ------ Joint test ------
    Variable |        Obs   Pr(skewness)   Pr(kurtosis)   Adj chi2(2)  Prob>chi2
-------------+-------------------------------------------------------------------
         age |    109,546         0.0000         0.0000             .          .

Skewness and kurtosis tests for normality
                                                       ------ Joint test ------
    Variable |        Obs   Pr(skewness)   Pr(kurtosis)   Adj chi2(2)  Prob>chi2
-------------+-------------------------------------------------------------------
       thinc |    109,546         0.0000         0.0000             .          .

Skewness and kurtosis tests for normality
                                                       ------ Joint test ------
    Variable |        Obs   Pr(skewness)   Pr(kurtosis)   Adj chi2(2)  Prob>chi2
-------------+-------------------------------------------------------------------
        yedu |    109,546         0.0000         0.0000        106.40     0.0000
(11,207 real changes made)

Summary for variables: id
Group variable: htype (Household type)

           htype |         N
-----------------+----------
             1 S |     37717
          1 CNRP |      8553
           1 C2R |     62737
    1 CNRP + 1 S |        17
         Multi S |       372
    1 C2R  + 1 S |       117
  1 C2R + 1 CNRP |         2
       Multi C2R |        29
 Multi S + 1 C2R |         2
-----------------+----------
           Total |    109546
-----------------------------------------------------------------------------

  Household
       type |      Freq.     Percent        Cum.
------------+-----------------------------------
          1 |     37,717       34.43       34.43
          2 |      8,553        7.81       42.24
          3 |     62,737       57.27       99.51
          4 |         17        0.02       99.52
          5 |        372        0.34       99.86
          6 |        117        0.11       99.97
          7 |          2        0.00       99.97
          8 |         29        0.03      100.00
         10 |          2        0.00      100.00
------------+-----------------------------------
      Total |    109,546      100.00
(37,717 real changes made)
(62,737 real changes made)
```

```
note: 2.naly omitted because of collinearity.
note: 2.rhfo#0.ghto identifies no observations in the sample.
note: 2.rhfo#1.ghto identifies no observations in the sample.
note: 2.rhfo#2.ghto omitted because of collinearity.
Iteration 0:  Log likelihood = -155895.31
Iteration 1:  Log likelihood = -129377.62
Iteration 2:  Log likelihood = -127777.18
Iteration 3:  Log likelihood = -127767.69
Iteration 4:  Log likelihood = -127767.68

Ordered logistic regression                         Number of obs =  109,546
                                                    LR chi2(37)   = 56255.24
                                                    Prob > chi2   =   0.0000
Log likelihood = -127767.68                         Pseudo R2     =   0.1804
```

sphus	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]	
age	.0894109	.0136668	6.54	0.000	.0626244	.1161974
c.age#c.age	-.0004176	.0000905	-4.62	0.000	-.0005949	-.0002403
yedu	-.006069	.0049411	-1.23	0.219	-.0157533	.0036153
c.yedu#c.yedu	-.0015132	.0002163	-6.99	0.000	-.0019372	-.0010892
thinc	-.1957495	.0138038	-14.18	0.000	-.2228043	-.1686946
c.thinc#c.thinc	.0260581	.00617	4.22	0.000	.0139651	.0381511
gender						
Female	-.0248661	.0126354	-1.97	0.049	-.0496311	-.0001011
1.covid	-.1181847	.0196721	-6.01	0.000	-.1567414	-.079628
household						
Couple, both responding	.1764051	.0152892	11.54	0.000	.1464388	.2063714
Multiple, at least 1 responding	.2046971	.0242875	8.43	0.000	.1570946	.2522997
cjs						
Not applicable	.5731575	.0586075	9.78	0.000	.458289	.688026
Retired	.3187207	.0295373	10.79	0.000	.2608287	.3766127
Unemployed	.2371021	.0879429	2.70	0.007	.0647373	.409467
Permanently sick or disabled	1.465681	.0628443	23.32	0.000	1.342508	1.588853
Homemaker	.3780172	.0362123	10.44	0.000	.3070425	.4489919
Other	.2272851	.0625474	3.63	0.000	.1046945	.3498758
gali						
Limited	1.994513	.0139072	143.42	0.000	1.967256	2.021771
lifesat	-.3113203	.0189781	-16.40	0.000	-.3485167	-.2741239
c.lifesat#c.lifesat	-.000214	.0013209	-0.16	0.871	-.0028029	

```
>     .0023748
                                  |
                            naly  |
                     At least 1   |  -.3682765   .0190147   -19.37   0.000    -.4055446
>    -.3310084
                  Not applicable  |          0  (omitted)
                                  |
                            rhfo  |
                     At least 1   |   .4022668   .0184699    21.78   0.000     .3660664
>     .4384672
                  Not applicable  |  -.1768136   .0325619    -5.43   0.000    -.2406338
>    -.1129934
                                  |
                            ghto  |
                     At least 1   |   -.234468   .0181715   -12.90   0.000    -.2700834
>    -.1988526
                  Not applicable  |   .0230997   .0197823     1.17   0.243     -.015673
>     .0618723
                                  |
                       rhfo#ghto  |
         At least 1#At least 1    |  -.1949625   .0334822    -5.82   0.000    -.2605864
>    -.1293386
      At least 1#Not applicable   |  -.0299268    .046156    -0.65   0.517    -.1203908
>     .0605372
           Not applicable#None    |          0  (empty)
      Not applicable#At least 1   |          0  (empty)
  Not applicable#Not applicable   |          0  (omitted)
                                  |
                         country  |
                         Austria  |   .1860174   .0296509     6.27   0.000     .1279026
>     .2441322
                         Germany  |   .6591807    .036529    18.05   0.000     .5875852
>     .7307762
                     Netherlands  |   .0750803   .0359947     2.09   0.037      .004532
>     .1456286
                           Spain  |   .7712743   .0337807    22.83   0.000     .7050653
>     .8374833
                           Italy  |   .5764137     .03276    17.60   0.000     .5122053
>     .6406221
                          France  |   .4966919    .029578    16.79   0.000     .4387201
>     .5546638
                         Denmark  |  -.2588955   .0348729    -7.42   0.000    -.3272451
>     -.190546
                     Switzerland  |   .0673866   .0320368     2.10   0.035     .0045956
>     .1301775
                         Belgium  |   .1172657   .0296127     3.96   0.000     .0592258
>     .1753056
                  Czech Republic  |   .4216311   .0319127    13.21   0.000     .3590834
>     .4841788
                        Slovenia  |   .5907589   .0349022    16.93   0.000     .5223517
>      .659166
------------------------------------+----------------------------------------------------------------
  ------------
                           /cut1  |  -.5546385   .5124182                      -1.55896
>     .4496827
                           /cut2  |   1.224105   .5125127                       .2195981
>     2.228611
                           /cut3  |   3.712274   .5127701                       2.707263
>     4.717285
                           /cut4  |   6.006924   .5127537                       5.001946
>     7.011903
------------------------------------+----------------------------------------------------------------
  ------------
```

Tests of proportional odds assumption

```
Number of obs            =  109,546
Number of predictors     =      37
Number of outcome levels =       5
Degrees of freedom       =     111
```

Test	chi2	P>chi2
Brant	**4061.55**	**0.000**
Likelihood-ratio	**4006.81**	**0.000**
Score	**4105.82**	**0.000**
Wald	**4127.41**	**0.000**
Wolfe-Gould	**4045.56**	**0.000**

(file **TestAssum_4.doc** not found)
Click to Open File: TestAssum_4.doc
note: **2.naly** omitted because of collinearity.
note: **2.rhfo#0.ghto** identifies no observations in the sample.
note: **2.rhfo#1.ghto** identifies no observations in the sample.
note: **2.rhfo#2.ghto** omitted because of collinearity.

Generalized Ordered Logit Estimates Number of obs = **109,528**
Wald chi2(1) = .
Prob > chi2 = .
Log pseudolikelihood = **-122307.27** Pseudo R2 = **0.1985**

(Std. err. adjusted for **6** cluste
> rs in **Qyear**)

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% con
Excellent					
age	**.103928**	**.0836269**	**1.24**	**0.214**	**-.0599777**
>	**.2678336**				
c.age#c.age	**-.0004909**	**.0005929**	**-0.83**	**0.408**	**-.0016529**
>	**.0006711**				
yedu	**.0242934**	**.0229839**	**1.06**	**0.291**	**-.0207543**
>	**.069341**				
c.yedu#c.yedu	**-.0018433**	**.0010228**	**-1.80**	**0.072**	**-.003848**
>	**.0001614**				
thinc	**-.1132283**	**.05531**	**-2.05**	**0.041**	**-.2216338**
>	**-.0048227**				
c.thinc#c.thinc	**.0009316**	**.0293157**	**0.03**	**0.975**	**-.0565262**
>	**.0583894**				
gender					
Female	**.139266**	**.059034**	**2.36**	**0.018**	**.0235614**
>	**.2549706**				
covid					
1	**-.0000699**	**.0125176**	**-0.01**	**0.996**	**-.0246039**
>	**.0244642**				
household					
Couple, both responding	**.1670126**	**.0550533**	**3.03**	**0.002**	**.0591102**
>	**.274915**				
Multiple, at least 1 responding	**.1729861**	**.0756956**	**2.29**	**0.022**	**.0246254**
>	**.3213468**				
cjs					
Not applicable	**.3293641**	**.5442349**	**0.61**	**0.545**	**-.7373167**
>	**1.396045**				
Retired	**.2866301**	**.1201592**	**2.39**	**0.017**	**.0511225**
>	**.5221378**				
Unemployed	**.1085894**	**.2682576**	**0.40**	**0.686**	**-.4171858**
>	**.6343647**				
Permanently sick or disabled	**.6362609**	**.1322262**	**4.81**	**0.000**	**.3771023**
>	**.8954195**				
Homemaker	**.374294**	**.1986646**	**1.88**	**0.060**	**-.0150814**
>	**.7636693**				

```
                        Other |  -.3064439   .3443685    -0.89   0.374    -.9813936
>     .3685059
                              |
                         gali |
                      Limited |   2.033054    .042771    47.53   0.000     1.949224
>     2.116883
                              |
                      lifesat |   .4810263   .0876274     5.49   0.000     .3092798
>     .6527728
                              |
            c.lifesat#c.lifesat |  -.0603789   .0056088   -10.76   0.000     -.071372
>    -.0493858
                              |
                         naly |
                    At least 1 |  -.1727672   .0154751   -11.16   0.000    -.2030978
>    -.1424365
                              |
                         rhfo |
                    At least 1 |   .2015491    .063768     3.16   0.002      .076566
>     .3265321
                Not applicable |  -.2175201   .0550257    -3.95   0.000    -.3253685
>    -.1096718
                              |
                         ghto |
                    At least 1 |   -.193014   .0947766    -2.04   0.042    -.3787726
>    -.0072553
                Not applicable |  -.0506809   .0260142    -1.95   0.051    -.1016678
>     .0003061
                              |
                     rhfo#ghto |
        At least 1#At least 1 |  -.1970925   .0938557    -2.10   0.036    -.3810463
>    -.0131388
     At least 1#Not applicable |  -.1705074   .0691048    -2.47   0.014    -.3059503
>    -.0350645
                              |
                      country |
                      Austria |   .7149518   .0826033     8.66   0.000     .5530523
>     .8768512
                      Germany |   1.371523   .1166653    11.76   0.000     1.142863
>     1.600183
                  Netherlands |   -.121033   .1012149    -1.20   0.232    -.3194106
>     .0773447
                        Spain |   1.512559   .0549008    27.55   0.000     1.404956
>     1.620163
                        Italy |   .6493993   .0796481     8.15   0.000      .493292
>     .8055067
                       France |   .6776742   .0427424    15.85   0.000     .5939007
>     .7614478
                      Denmark |   .0480693   .0616308     0.78   0.435    -.0727248
>     .1688635
                  Switzerland |   .8310128   .1038002     8.01   0.000     .6275682
>     1.034457
                      Belgium |   .6686923   .0546027    12.25   0.000      .561673
>     .7757117
                Czech Republic |   1.407741   .1656596     8.50   0.000     1.083055
>     1.732428
                     Slovenia |   1.046763   .1972226     5.31   0.000     .6602137
>     1.433312
                              |
                        _cons |  -3.257655    3.22294    -1.01   0.312    -9.574501
>     3.059191
-------------------------------+----------------------------------------------------------
  ____________
Very_good                     |
                          age |   .0436922     .03157     1.38   0.166    -.0181838
>     .1055682
                              |
                   c.age#c.age |   -.000067   .0002057    -0.33   0.745    -.0004701
>     .0003362
                              |
                         yedu |   .0057881   .0168057     0.34   0.731    -.0271505
>     .0387267
```

```
                                 |
                  c.yedu#c.yedu  |   -.002376   .0007208    -3.30   0.001    -.0037887
>    -.0009633
                                 |
                          thinc  |  -.2042554   .0610806    -3.34   0.001    -.3239713
>    -.0845396
                                 |
              c.thinc#c.thinc    |   .0161254   .0237544     0.68   0.497    -.0304323
>     .0626832
                                 |
                         gender  |
                         Female  |   .0641421   .0238119     2.69   0.007     .0174717
>     .1108126
                                 |
                          covid  |
                              1  |  -.0573331   .0288208    -1.99   0.047    -.1138208
>    -.0008453
                                 |
                      household  |
        Couple, both responding  |    .232297   .0287668     8.08   0.000     .1759152
>     .2886788
Multiple, at least 1 responding  |   .2684942   .0456239     5.88   0.000      .179073
>     .3579155
                                 |
                            cjs  |
                Not applicable   |   .2615615   .3899298     0.67   0.502    -.5026868
>      1.02581
                        Retired  |    .338784   .0627238     5.40   0.000     .2158476
>     .4617205
                     Unemployed  |   .2075585   .1365889     1.52   0.129    -.0601509
>     .4752679
   Permanently sick or disabled  |   .6423658   .1440656     4.46   0.000     .3600024
>     .9247292
                      Homemaker  |   .2879134   .0502824     5.73   0.000     .1893618
>     .3864651
                          Other  |  -.1785401    .345482    -0.52   0.605    -.8556724
>     .4985921
                                 |
                           gali  |
                        Limited  |    1.83578   .0314622    58.35   0.000     1.774115
>     1.897445
                                 |
                         lifesat |   .0488687   .0918725     0.53   0.595     -.131198
>     .2289355
                                 |
            c.lifesat#c.lifesat  |   -.026668   .0059657    -4.47   0.000    -.0383606
>    -.0149754
                                 |
                           naly  |
                      At least 1 |   .0286853   .0446145     0.64   0.520    -.0587575
>     .1161281
                                 |
                           rhfo  |
                      At least 1 |   .1716835   .0455448     3.77   0.000     .0824173
>     .2609496
                Not applicable   |  -.1886715   .0696431    -2.71   0.007    -.3251695
>    -.0521735
                                 |
                           ghto  |
                      At least 1 |  -.1428215   .0334053    -4.28   0.000    -.2082947
>    -.0773483
                Not applicable   |  -.0122878   .0380907    -0.32   0.747    -.0869442
>     .0623686
                                 |
                       rhfo#ghto |
          At least 1#At least 1  |  -.0929945   .1469982    -0.63   0.527    -.3811057
>     .1951167
       At least 1#Not applicable |  -.0816857   .0548949    -1.49   0.137    -.1892778
>     .0259063
                                 |
                         country |
                         Austria |   .2436276   .0402745     6.05   0.000     .1646909
```

```
>     .3225642
                    Germany |   1.103071    .065327    16.89   0.000     .9750324
>     1.231109
                Netherlands |    .381952    .056124     6.81   0.000      .271951
>      .491953
                      Spain |   1.047962   .0441803    23.72   0.000       .96137
>     1.134553
                      Italy |   .8964851   .0464382    19.30   0.000     .8054678
>     .9875023
                     France |   .8683763   .0321399    27.02   0.000     .8053832
>     .9313693
                    Denmark |  -.3649129   .0601274    -6.07   0.000    -.4827605
>    -.2470653
                Switzerland |   .3736312   .0376155     9.93   0.000     .2999061
>     .4473563
                    Belgium |   .4985186    .052571     9.48   0.000     .3954814
>     .6015557
            Czech Republic  |   .9261187   .0692026    13.38   0.000      .790484
>     1.061753
                   Slovenia |   .9929769   .1150852     8.63   0.000     .7674141
>      1.21854
                            |
                      _cons |  -1.542672   1.148081    -1.34   0.179    -3.792869
>     .7075246
------------------------------+----------------------------------------------------
  ___________
Good                        |
                        age |   .1441218   .0257212     5.60   0.000     .0937091
>     .1945344
                            |
                 c.age#c.age |  -.0008102   .0001703    -4.76   0.000    -.0011441
>    -.0004764
                            |
                       yedu |  -.0343666   .0051253    -6.71   0.000    -.0444119
>    -.0243213
                            |
              c.yedu#c.yedu |  -.0004414   .0003115    -1.42   0.156    -.0010518
>     .0001691
                            |
                      thinc |  -.2265777    .029214    -7.76   0.000     -.283836
>    -.1693193
                            |
            c.thinc#c.thinc |   .0570882   .0176199     3.24   0.001     .0225539
>     .0916226
                            |
                     gender |
                     Female |  -.0172882   .0181824    -0.95   0.342     -.052925
>     .0183485
                            |
                      covid |
                          1 |  -.1039378   .0224748    -4.62   0.000    -.1479876
>    -.0598879
                            |
                  household |
        Couple, both responding |   .1484659   .0408843     3.63   0.000     .0683342
>     .2285976
Multiple, at least 1 responding |   .2579512   .0532054     4.85   0.000     .1536706
>     .3622319
                            |
                        cjs |
            Not applicable  |   .4228882   .2524533     1.68   0.094    -.0719112
>     .9176876
                    Retired |   .2760298   .1151294     2.40   0.017     .0503804
>     .5016792
                 Unemployed |   .1344098   .1189772     1.13   0.259    -.0987813
>     .3676009
   Permanently sick or disabled |   1.045546   .0817407    12.79   0.000     .8853371
>     1.205755
                  Homemaker |    .293232   .2008339     1.46   0.144    -.1003953
>     .6868593
                      Other |   .0097383   .2673883     0.04   0.971    -.5143331
>     .5338097
```

```
                                 |
                        gali     |
                     Limited     |   2.114133   .0158527   133.36   0.000     2.083062
>     2.145204
                                 |
                     lifesat     |  -.2831132   .0628982    -4.50   0.000    -.4063914
>    -.1598349
                                 |
           c.lifesat#c.lifesat   |   -.001383    .004219    -0.33   0.743    -.0096521
>     .0068861
                                 |
                        naly     |
                  At least 1     |  -.2112062   .0508332    -4.15   0.000    -.3108374
>     -.111575
                                 |
                        rhfo     |
                  At least 1     |   .3984886   .0642437     6.20   0.000     .2725733
>     .5244038
              Not applicable     |  -.1675845   .0423178    -3.96   0.000    -.2505259
>    -.0846431
                                 |
                        ghto     |
                  At least 1     |  -.2392239   .0581886    -4.11   0.000    -.3532715
>    -.1251763
              Not applicable     |  -.0050599   .0264712    -0.19   0.848    -.0569425
>     .0468227
                                 |
                    rhfo#ghto    |
       At least 1#At least 1     |  -.1238058   .1431776    -0.86   0.387    -.4044288
>     .1568171
     At least 1#Not applicable   |  -.0017279   .0700936    -0.02   0.980    -.1391088
>     .1356531
                                 |
                     country     |
                     Austria     |  -.0393289   .0979719    -0.40   0.688    -.2313503
>     .1526924
                     Germany     |   .3053263   .0726116     4.20   0.000     .1630102
>     .4476423
                 Netherlands     |  -.1199148   .0353866    -3.39   0.001    -.1892713
>    -.0505583
                       Spain     |   .5635886   .0658074     8.56   0.000     .4346084
>     .6925688
                       Italy     |   .5232701   .1110839     4.71   0.000     .3055497
>     .7409905
                      France     |   .1174808   .0493343     2.38   0.017     .0207874
>     .2141742
                     Denmark     |  -.0656207   .0945701    -0.69   0.488    -.2509746
>     .1197332
                 Switzerland     |  -.3955437   .0732688    -5.40   0.000     -.539148
>    -.2519395
                     Belgium     |  -.2760948    .034128    -8.09   0.000    -.3429845
>    -.2092051
              Czech Republic     |  -.1583857   .1304366    -1.21   0.225    -.4140368
>     .0972655
                    Slovenia     |   .1781203   .0850037     2.10   0.036     .0115161
>     .3447244
                                 |
                       _cons     |   -5.48791   .8839902    -6.21   0.000    -7.220499
>    -3.755321
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  ____________
Fair                             |
                         age     |   .1748444   .0420561     4.16   0.000      .092416
>     .2572729
                                 |
                   c.age#c.age   |  -.0010328     .00028    -3.69   0.000    -.0015816
>     -.000484
                                 |
                        yedu     |  -.0240004   .0039351    -6.10   0.000     -.031713
>    -.0162879
                                 |
                  c.yedu#c.yedu  |  -.0005515   .0002638    -2.09   0.037    -.0010686
```

```
>    -.0000345
                                  |
                          thinc   |  -.1688639   .0716259    -2.36   0.018    -.3092481
>    -.0284798
                                  |
                c.thinc#c.thinc   |   .0378773   .0345274     1.10   0.273    -.0297952
>     .1055498
                                  |
                         gender   |
                         Female   |   -.079199   .0514887    -1.54   0.124    -.1801149
>     .0217169
                                  |
                          covid   |
                              1   |  -.1124312   .0481101    -2.34   0.019    -.2067252
>    -.0181373
                                  |
                      household   |
        Couple, both responding   |   .1185572    .055277     2.14   0.032     .0102162
>     .2268982
Multiple, at least 1 responding   |   .0911346    .068821     1.32   0.185    -.0437521
>     .2260213
                                  |
                            cjs   |
                Not applicable    |   .4515647   .1710951     2.64   0.008     .1162245
>     .7869049
                        Retired   |   .2844185   .0796365     3.57   0.000     .1283339
>     .4405031
                     Unemployed   |   -.723911   .1371009    -5.28   0.000    -.9926238
>    -.4551981
   Permanently sick or disabled   |   1.460897   .0944714    15.46   0.000     1.275736
>     1.646057
                      Homemaker   |   .2430795   .0822308     2.96   0.003     .0819102
>     .4042489
                          Other   |   .1045453   .1460605     0.72   0.474    -.1817279
>     .3908186
                                  |
                           gali   |
                        Limited   |   2.998625   .0680748    44.05   0.000     2.865201
>     3.132049
                                  |
                         lifesat  |  -.5613585   .0485334   -11.57   0.000    -.6564822
>    -.4662349
                                  |
             c.lifesat#c.lifesat  |   .0201982    .003604     5.60   0.000     .0131344
>     .0272619
                                  |
                           naly   |
                     At least 1   |  -.5373176   .0859449    -6.25   0.000    -.7057665
>    -.3688686
                                  |
                           rhfo   |
                     At least 1   |   .5772664   .0323028    17.87   0.000      .513954
>     .6405787
                Not applicable    |  -.1592207   .1050551    -1.52   0.130    -.3651249
>     .0466835
                                  |
                           ghto   |
                     At least 1   |  -.5789038    .070499    -8.21   0.000    -.7170793
>    -.4407284
                Not applicable    |   .1498568   .0261451     5.73   0.000     .0986133
>     .2011003
                                  |
                      rhfo#ghto   |
          At least 1#At least 1   |  -.0025046    .160413    -0.02   0.988    -.3169084
>     .3118992
      At least 1#Not applicable   |   .0627664   .0304621     2.06   0.039     .0030617
>     .1224711
                                  |
                         country  |
                         Austria  |  -.1178737   .2033885    -0.58   0.562    -.5165078
>     .2807605
                         Germany  |   .0544075   .1661173     0.33   0.743    -.2711765
```

```
>     .3799914
                  Netherlands |  -.434306   .149514   -2.90   0.004   -.7273481
>    -.1412639
                        Spain |  .4721939  .1576079    3.00   0.003    .1632882
>     .7810997
                        Italy | -.0714528  .1393267   -0.51   0.608   -.3445282
>     .2016225
                       France |  .2387447  .1599477    1.49   0.136   -.0747471
>     .5522365
                      Denmark |  .0022148   .197322    0.01   0.991   -.3845292
>     .3889589
                  Switzerland | -.4624603  .1719799   -2.69   0.007   -.7995347
>    -.1253859
                      Belgium | -.4784295   .082526   -5.80   0.000   -.6401774
>    -.3166815
              Czech Republic  | -.0184607  .2026454   -0.09   0.927   -.4156384
>      .378717
                     Slovenia |  .3645995  .1311671    2.78   0.005    .1075167
>     .6216824
                              |
                        _cons | -8.568511  1.657696   -5.17   0.000   -11.81754
>    -5.319487
------------------------------+----------------------------------------------------
  ____________

WARNING! 9 in-sample cases have an outcome with a predicted probability that is
less than 0. See the gologit2 help section on Warning Messages for more information.
results_imp1_go_4.xls
dir : seeout
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -2.890e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.861e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.345e+08  (not concave)
Iteration 3:  Log pseudolikelihood = -1.125e+08  (not concave)
Iteration 4:  Log pseudolikelihood =  -92399699  (not concave)
Iteration 5:  Log pseudolikelihood =  -87640616  (not concave)
Iteration 6:  Log pseudolikelihood =  -84469171  (not concave)
Iteration 7:  Log pseudolikelihood =  -82857899  (not concave)
Iteration 8:  Log pseudolikelihood =  -81438518  (not concave)
Iteration 9:  Log pseudolikelihood =  -80936504  (not concave)
Iteration 10: Log pseudolikelihood =  -80668820  (not concave)
Iteration 11: Log pseudolikelihood =  -80389154  (not concave)
Iteration 12: Log pseudolikelihood =  -79804453  (not concave)
Iteration 13: Log pseudolikelihood =  -79591242  (not concave)
Iteration 14: Log pseudolikelihood =  -79465729  (not concave)
Iteration 15: Log pseudolikelihood =  -79295465  (not concave)
Iteration 16: Log pseudolikelihood =  -79198261  (not concave)
Iteration 17: Log pseudolikelihood =  -79072599  (not concave)
Iteration 18: Log pseudolikelihood =  -78980056  (not concave)
Iteration 19: Log pseudolikelihood =  -78892131
Iteration 20: Log pseudolikelihood =  -78250258  (not concave)
Iteration 21: Log pseudolikelihood =  -77878388  (not concave)
Iteration 22: Log pseudolikelihood =  -77830569
Iteration 23: Log pseudolikelihood =  -77774953  (not concave)
Iteration 24: Log pseudolikelihood =  -77745779
Iteration 25: Log pseudolikelihood =  -77736382
Iteration 26: Log pseudolikelihood =  -77733462
Iteration 27: Log pseudolikelihood =  -77733423
Iteration 28: Log pseudolikelihood =  -77733423

Stereotype logistic regression                      Number of obs =  29,989
                                                    Wald chi2(35) = 2225.89
Log pseudolikelihood = -77733423                    Prob > chi2   =  0.0000
```

(1) **[phi1_1]_cons = 1**

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.2784046	.1292409	2.15	0.031	.0250972	.5317121
c.age#c.age	-.0014432	.0008865	-1.63	0.104	-.0031807	.0002944
yedu	-.0135705	.0390514	-0.35	0.728	-.0901098	.0629687
c.yedu#c.yedu	-.0048289	.0018491	-2.61	0.009	-.008453	-.0012047
thinc	-.6081852	.1130365	-5.38	0.000	-.8297328	-.3866377
c.thinc#c.thinc	.1099575	.0520619	2.11	0.035	.007918	.211997
gender						
Female	.0435799	.1075772	0.41	0.685	-.1672676	.2544273
0.covid	0	(omitted)				
household						
Couple, both responding	.4041623	.1422779	2.84	0.005	.1253028	.6830219
Multiple, at least 1 responding	.7951109	.1943419	4.09	0.000	.4142078	1.176014
cjs						
Not applicable	.0951008	.9017166	0.11	0.916	-1.672231	1.862433
Retired	.6280014	.2088716	3.01	0.003	.2186207	1.037382
Unemployed	-.2467041	.4121271	-0.60	0.549	-1.054458	.5610502
Permanently sick or disabled	2.976235	.4532027	6.57	0.000	2.087974	3.864496
Homemaker	.3850127	.243043	1.58	0.113	-.0913428	.8613682
Other	-1.779513	.4840704	-3.68	0.000	-2.728274	-.8307523
gali						
Limited	5.378017	.1551183	34.67	0.000	5.07399	5.682043
lifesat	-.5551819	.1921469	-2.89	0.004	-.931783	-.1785809
c.lifesat#c.lifesat	-.0147768	.0134488	-1.10	0.272	-.0411359	.0115823
naly						
At least 1	-.9148566	.1397814	-6.54	0.000	-1.188823	-.6408901
Not applicable	0	(omitted)				
rhfo						
At least 1	1.214901	.1777206	6.84	0.000	.8665754	1.563227
ghto						
At least 1	-.4604069	.1535803	-3.00	0.003	-.7614188	-.159395

```
                 Not applicable |  -.0352011   .1238683    -0.28   0.776    -.2779786
>     .2075763                  |
                                |
                      rhfo#ghto |
          At least 1#At least 1 |  -1.052283   .3336875    -3.15   0.002    -1.706299
>    -.3982679
      At least 1#Not applicable |   .0075981   .2902825     0.03   0.979    -.5613452
>     .5765414                  |
                                |
                        country |
                        Austria |  -.2625788   .1932829    -1.36   0.174    -.6414063
>     .1162486
                        Germany |   1.068444    .220213     4.85   0.000     .6368341
>     1.500053
                    Netherlands |  -.1331048   .2070943    -0.64   0.520    -.5390022
>     .2727926
                          Spain |   1.962658   .2239703     8.76   0.000     1.523684
>     2.401632
                          Italy |   1.062131   .2135929     4.97   0.000     .6434965
>     1.480765
                         France |   .9478306    .192871     4.91   0.000     .5698103
>     1.325851
                        Denmark |  -1.290206   .2402393    -5.37   0.000    -1.761066
>    -.8193457
                    Switzerland |  -.5512791   .2194635    -2.51   0.012    -.9814196
>    -.1211386
                        Belgium |   .0012407   .1932605     0.01   0.995    -.3775428
>     .3800243
                 Czech Republic |   .6591118   .2320448     2.84   0.005     .2043124
>     1.113911
                       Slovenia |   1.424166   .2736424     5.20   0.000     .8878369
>     1.960496
-------------------------------+----------------------------------------------------------
  ____________
                        /phi1_1 |          1  (constrained)
                        /phi1_2 |  .8535814   .0180443    47.30   0.000     .8182153
>     .8889475
                        /phi1_3 |  .6255516   .0134272    46.59   0.000     .5992348
>     .6518684
                        /phi1_4 |  .3437465   .0119391    28.79   0.000     .3203464
>     .3671467
                        /phi1_5 |          0  (base outcome)
-------------------------------+----------------------------------------------------------
  ____________
                        /theta1 |  9.844037    4.72152     2.08   0.037     .5900269
>     19.09805
                        /theta2 |  9.778931   4.035105     2.42   0.015      1.87027
>     17.68759
                        /theta3 |   8.98892   2.974319     3.02   0.003     3.159362
>     14.81848
                        /theta4 |  5.714196   1.652519     3.46   0.001     2.475317
>     8.953074
                        /theta5 |          0  (base outcome)
-------------------------------+----------------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_11_4.xls
dir : seeout

Predictive margins                                      Number of obs = 29,989
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))
```

	Margin	Delta-method std. err.	z	P>|z|	[95% conf. interval]
_predict#rhfo#ghto					
1#None#None	.0529461	.0021254	24.91	0.000	.0487803 .0571119
1#None#At least 1	.060903	.0030127	20.22	0.000	.0549981 .0668078
1#None#Not applicable	.0535267	.0023955	22.35	0.000	.0488317 .0582218
1#At least 1#None	.0355757	.0025222	14.10	0.000	.0306322 .0405191
1#At least 1#At least 1	.0580022	.0049663	11.68	0.000	.0482685 .0677359
1#At least 1#Not applicable	.0359153	.0031577	11.37	0.000	.0297263 .0421043
2#None#None	.132695	.0034119	38.89	0.000	.1260078 .1393822
2#None#At least 1	.1450809	.0047267	30.69	0.000	.1358169 .154345
2#None#Not applicable	.1336289	.0038269	34.92	0.000	.1261283 .1411295
2#At least 1#None	.102111	.0046168	22.12	0.000	.0930623 .1111598
2#At least 1#At least 1	.1406653	.0075245	18.69	0.000	.1259175 .1554132
2#At least 1#Not applicable	.102766	.005882	17.47	0.000	.0912375 .1142945
3#None#None	.3890645	.0048895	79.57	0.000	.3794812 .3986478
3#None#At least 1	.3982304	.0054308	73.33	0.000	.3875861 .4088747
3#None#Not applicable	.3898223	.0051508	75.68	0.000	.379727 .3999177
3#At least 1#None	.3576152	.0061891	57.78	0.000	.3454848 .3697456
3#At least 1#At least 1	.3951798	.0068027	58.09	0.000	.3818467 .4085128
3#At least 1#Not applicable	.3584353	.0078608	45.60	0.000	.3430283 .3738422
4#None#None	.3095006	.0046446	66.64	0.000	.3003973 .3186039
4#None#At least 1	.2957095	.0057042	51.84	0.000	.2845295 .3068896
4#None#Not applicable	.3084842	.0050592	60.97	0.000	.2985683 .3184001
4#At least 1#None	.3400954	.006311	53.89	0.000	.327726 .3524647
4#At least 1#At least 1	.3007001	.0088385	34.02	0.000	.283377 .3180233
4#At least 1#Not applicable	.3395055	.0069286	49.00	0.000	.3259256 .3530854
5#None#None	.1157937	.0036535	31.69	0.000	.1086329 .1229545
5#None#At least 1	.1000761	.0050169	19.95	0.000	.0902433 .109909
5#None#Not applicable	.1145378	.0043358	26.42	0.000	.1060398 .1230358
5#At least 1#None	.1646027	.0071466	23.03	0.000	.1505957 .1786097
5#At least 1#At least 1	.1054526	.0088902	11.86	0.000	.0880281 .122877
5#At least 1#Not applicable	.1633779	.0103121	15.84	0.000	.1431665 .1835892

```
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -2.394e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -2.009e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.142e+08  (not concave)
Iteration 3:  Log pseudolikelihood =  -88539996  (not concave)
Iteration 4:  Log pseudolikelihood =  -69560072  (not concave)
Iteration 5:  Log pseudolikelihood =  -64646992  (not concave)
Iteration 6:  Log pseudolikelihood =  -62779221  (not concave)
Iteration 7:  Log pseudolikelihood =  -61649848  (not concave)
Iteration 8:  Log pseudolikelihood =  -60675981  (not concave)
Iteration 9:  Log pseudolikelihood =  -59796158  (not concave)
Iteration 10: Log pseudolikelihood =  -59513829  (not concave)
Iteration 11: Log pseudolikelihood =  -59359880  (not concave)
Iteration 12: Log pseudolikelihood =  -59202881  (not concave)
Iteration 13: Log pseudolikelihood =  -58613502  (not concave)
Iteration 14: Log pseudolikelihood =  -58528358  (not concave)
Iteration 15: Log pseudolikelihood =  -58477603  (not concave)
Iteration 16: Log pseudolikelihood =  -58429569  (not concave)
Iteration 17: Log pseudolikelihood =  -58319322  (not concave)
Iteration 18: Log pseudolikelihood =  -58162128  (not concave)
Iteration 19: Log pseudolikelihood =  -58123400  (not concave)
Iteration 20: Log pseudolikelihood =  -58091576  (not concave)
Iteration 21: Log pseudolikelihood =  -58061003
Iteration 22: Log pseudolikelihood =  -57491552  (backed up)
Iteration 23: Log pseudolikelihood =  -57378311  (not concave)
Iteration 24: Log pseudolikelihood =  -57275332
Iteration 25: Log pseudolikelihood =  -57254412
Iteration 26: Log pseudolikelihood =  -57248395
Iteration 27: Log pseudolikelihood =  -57248035
Iteration 28: Log pseudolikelihood =  -57248035

Stereotype logistic regression                        Number of obs =  22,899
                                                      Wald chi2(35) = 1186.77
Log pseudolikelihood = -57248035                      Prob > chi2   =  0.0000

 ( 1)  [phi1_1]_cons = 1
```

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.4374078	.1580873	2.77	0.006	.1275623	.7472533
c.age#c.age	-.0025975	.0010611	-2.45	0.014	-.0046773	-.0005178
yedu	-.0919084	.0452615	-2.03	0.042	-.1806192	-.0031976
c.yedu#c.yedu	-.0006598	.0020049	-0.33	0.742	-.0045893	.0032696
thinc	-.6442657	.1345523	-4.79	0.000	-.9079835	-.380548
c.thinc#c.thinc	.1737591	.0637094	2.73	0.006	.048891	.2986271
gender						
Female	.0368192	.1237042	0.30	0.766	-.2056365	.2792749
0.covid	0	(omitted)				
household						
Couple, both responding	.4167592	.1560139	2.67	0.008	.1109775	.7225409
Multiple, at least 1 responding	.4030453	.2383636	1.69	0.091	-.0641388	

```
>     .8702293
                                  |
                            cjs   |
                Not applicable    |   1.667007   .7965561     2.09   0.036     .1057862
>     3.228229
                        Retired   |   .9481749   .2846922     3.33   0.001     .3901885
>     1.506161
                     Unemployed   |      .3996   .6636035     0.60   0.547    -.9010391
>     1.700239
   Permanently sick or disabled   |   3.418903   .5721862     5.98   0.000     2.297439
>     4.540368
                      Homemaker   |   1.129865   .3341511     3.38   0.001     .4749406
>     1.784789
                          Other   |   .1871958   .5311321     0.35   0.725    -.8538038
>     1.228196
                                  |
                           gali   |
                        Limited   |   5.079077   .1989044    25.54   0.000     4.689232
>     5.468923
                         lifesat  |  -.6033318    .180425    -3.34   0.001    -.9569584
>    -.2497053
                                  |
             c.lifesat#c.lifesat  |   -.008052   .0130072    -0.62   0.536    -.0335455
>     .0174416
                                  |
                           naly   |
                     At least 1   |  -.7318102   .1606976    -4.55   0.000    -1.046772
>    -.4168487
                Not applicable    |         0  (omitted)
                                  |
                           rhfo   |
                     At least 1   |   1.082974   .1972458     5.49   0.000     .6963789
>     1.469568
                                  |
                           ghto   |
                     At least 1   |  -.8190159   .1836035    -4.46   0.000    -1.178872
>    -.4591596
                Not applicable    |  -.1233529   .1413986    -0.87   0.383    -.4004891
>     .1537833
                                  |
                       rhfo#ghto  |
          At least 1#At least 1   |  -.6658673    .410252    -1.62   0.105    -1.469946
>     .1382118
      At least 1#Not applicable   |   .0537855   .3519022     0.15   0.879    -.6359302
>     .7435012
                                  |
                         country  |
                         Austria  |   .7648661   .2146488     3.56   0.000     .3441621
>      1.18557
                         Germany  |   2.044074   .2538844     8.05   0.000      1.54647
>     2.541678
                     Netherlands  |   .5417756   .2259291     2.40   0.016     .0989628
>     .9845885
                           Spain  |   2.400173   .2613649     9.18   0.000     1.887907
>     2.912439
                           Italy  |    1.81864   .2471613     7.36   0.000     1.334213
>     2.303067
                          France  |   1.780266   .2154877     8.26   0.000     1.357918
>     2.202614
                         Denmark  |  -.2028971   .2571123    -0.79   0.430    -.7068279
>     .3010337
                     Switzerland  |   .2790932   .2385505     1.17   0.242    -.1884572
>     .7466436
                         Belgium  |   .5441359   .2073277     2.62   0.009      .137781
>     .9504908
                Czech Republic    |   1.630058   .2733072     5.96   0.000     1.094386
>     2.165731
                        Slovenia  |   1.480035   .3048712     4.85   0.000     .8824983
>     2.077572
------------------------------------------------------------------------------------------
                          /phi1_1 |          1  (constrained)
```

```
                      /phi1_2 |   .8480733   .0264615    32.05   0.000     .7962096
>      .899937
                      /phi1_3 |   .6260206   .0199205    31.43   0.000     .5869771
>     .6650642
                      /phi1_4 |    .333639   .0155973    21.39   0.000     .3030689
>     .3642091
                      /phi1_5 |          0  (base outcome)
------------------------------+----------------------------------------------------
                      /theta1 |  16.00033   5.848304     2.74   0.006     4.537862
>     27.46279
                      /theta2 |  14.92964   4.989304     2.99   0.003     5.150785
>      24.7085
                      /theta3 |  12.84031   3.706948     3.46   0.001     5.574827
>      20.1058
                      /theta4 |  7.608188   2.004526     3.80   0.000     3.679389
>     11.53699
                      /theta5 |          0  (base outcome)
------------------------------+----------------------------------------------------
(sphus=Poor is the base outcome)
results_imp1_ste_13_4.xls
dir : seeout

Predictive margins                                      Number of obs = 22,899
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------+----------------------------------------------------
                              |            Delta-method
                              |     Margin   std. err.      z    P>|z|     [95% conf. i
> nterval]
------------------------------+----------------------------------------------------
          _predict#rhfo#ghto  |
                 1#None#None  |   .0468058   .0022415    20.88   0.000     .0424125
>  .051199
          1#None#At least 1   |   .0606238   .0036592    16.57   0.000     .0534519
> .0677956
      1#None#Not applicable   |   .0487236   .0025679    18.97   0.000     .0436907
> .0537566
          1#At least 1#None   |   .0322823   .0026933    11.99   0.000     .0270035
> .0375611
    1#At least 1#At least 1   |   .0532646   .0058027     9.18   0.000     .0418915
> .0646377
1#At least 1#Not applicable   |   .0330965   .0037433     8.84   0.000     .0257597
> .0404332
                 2#None#None  |   .1217233   .0038227    31.84   0.000     .1142308
> .1292157
          2#None#At least 1   |   .1433426   .0053588    26.75   0.000     .1328396
> .1538456
      2#None#Not applicable   |   .1248982   .0043731    28.56   0.000      .116327
> .1334693
          2#At least 1#None   |   .0953777   .0050242    18.98   0.000     .0855304
>  .105225
    2#At least 1#At least 1   |   .1321798   .0092069    14.36   0.000     .1141346
> .1502249
2#At least 1#Not applicable   |   .0969795   .0072346    13.41   0.000        .0828
>  .111159
                 3#None#None  |   .3919814   .0055477    70.66   0.000     .3811081
> .4028547
          3#None#At least 1   |    .410604   .0065785    62.42   0.000     .3977104
> .4234976
      3#None#Not applicable   |   .3951229   .0059085    66.87   0.000     .3835425
> .4067033
          3#At least 1#None   |   .3598527   .0074578    48.25   0.000     .3452356
```

```
> .3744698
    3#At least 1#At least 1 |   .4017877   .0091779    43.78   0.000     .3837993
> .4197761
3#At least 1#Not applicable |   .3621415   .0105509    34.32   0.000      .341462
>  .382821
                 4#None#None |   .3158889   .0054128    58.36   0.000       .30528
> .3264978
          4#None#At least 1 |    .290277   .0067929    42.73   0.000     .2769633
> .3035908
      4#None#Not applicable |   .3122568   .0057409    54.39   0.000     .3010048
> .3235087
          4#At least 1#None |   .3437059    .007319    46.96   0.000      .329361
> .3580508
    4#At least 1#At least 1 |   .3037471   .0111543    27.23   0.000     .2818851
> .3256092
4#At least 1#Not applicable |    .342156   .0087275    39.20   0.000     .3250504
> .3592617
                 5#None#None |   .1236006   .0042986    28.75   0.000     .1151756
> .1320257
          5#None#At least 1 |   .0951526   .0059304    16.04   0.000     .0835291
>  .106776
      5#None#Not applicable |   .1189985   .0052226    22.79   0.000     .1087624
> .1292346
          5#At least 1#None |   .1687815   .0081452    20.72   0.000     .1528172
> .1847457
    5#At least 1#At least 1 |   .1090208   .0116904     9.33   0.000     .0861079
> .1319336
5#At least 1#Not applicable |   .1656265   .0132145    12.53   0.000     .1397265
> .1915265
----------------------------+----------------------------------------------------------
  ________
slog13_4.xls
dir : seeout
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -1.695e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.177e+08  (not concave)
Iteration 2:  Log pseudolikelihood =  -64648978  (not concave)
Iteration 3:  Log pseudolikelihood =  -51044752  (not concave)
Iteration 4:  Log pseudolikelihood =  -48403707  (not concave)
Iteration 5:  Log pseudolikelihood =  -48023120  (not concave)
Iteration 6:  Log pseudolikelihood =  -47818476  (not concave)
Iteration 7:  Log pseudolikelihood =  -47624920
Iteration 8:  Log pseudolikelihood =  -47067968  (not concave)
Iteration 9:  Log pseudolikelihood =  -46943090
Iteration 10: Log pseudolikelihood =  -46875148  (not concave)
Iteration 11: Log pseudolikelihood =  -46659864
Iteration 12: Log pseudolikelihood =  -46580029  (not concave)
Iteration 13: Log pseudolikelihood =  -46533040
Iteration 14: Log pseudolikelihood =  -46517005
Iteration 15: Log pseudolikelihood =  -46512448
Iteration 16: Log pseudolikelihood =  -46512407
Iteration 17: Log pseudolikelihood =  -46512407

Stereotype logistic regression                        Number of obs =  18,545
                                                      Wald chi2(32) = 1120.05
Log pseudolikelihood = -46512407                      Prob > chi2   =  0.0000

 ( 1)  [phi1_1]_cons = 1
----------------------------+----------------------------------------------------------
  ____________
                            |               Robust
                      sphus | Coefficient  std. err.      z    P>|z|     [95% con
> f. interval]                |
----------------------------+----------------------------------------------------------
  ____________
                        age |   .5620139   .1716149     3.27   0.001     .2256548
>     .8983729
                            |
                 c.age#c.age |  -.0032802   .0011213    -2.93   0.003    -.0054779
>    -.0010825
```

```
                             |
                        yedu |  -.0683564   .0451173    -1.52   0.130    -.1567847
>      .020072
                             |
                c.yedu#c.yedu |   -.001402   .0020441    -0.69   0.493    -.0054084
>     .0026044
                             |
                       thinc |   -.444918   .1580187    -2.82   0.005    -.7546289
>     -.135207
                             |
             c.thinc#c.thinc |  -.0473563    .070391    -0.67   0.501    -.1853201
>     .0906075
                             |
                      gender |
                      Female |   .0708873    .139077     0.51   0.610    -.2016986
>     .3434732
                     0.covid |          0  (omitted)
                             |
                   household |
       Couple, both responding |   .3676396   .1589285     2.31   0.021     .0561455
>     .6791337
Multiple, at least 1 responding |   .6332887   .2671791     2.37   0.018     .1096272
>      1.15695
                             |
                         cjs |
             Not applicable |   2.974491   .8205922     3.62   0.000      1.36616
>     4.582822
                     Retired |   1.533268   .4542029     3.38   0.001     .6430471
>      2.42349
                  Unemployed |   .4531201   .8462513     0.54   0.592    -1.205502
>     2.111742
  Permanently sick or disabled |   3.365095   .7504137     4.48   0.000     1.894312
>     4.835879
                   Homemaker |   1.779366   .4928939     3.61   0.000      .813312
>     2.745421
                       Other |   1.691599   .5502589     3.07   0.002     .6131113
>     2.770087
                             |
                        gali |
                     Limited |   5.289927    .206553    25.61   0.000     4.885091
>     5.694764
                      lifesat |   -.904193   .1799232    -5.03   0.000    -1.256836
>      -.55155
                             |
          c.lifesat#c.lifesat |   .0073527   .0129945     0.57   0.572     -.018116
>     .0328213
                             |
                        naly |
                  At least 1 |  -.2952219   .1568413    -1.88   0.060    -.6026252
>     .0121814
             Not applicable |          0  (omitted)
                             |
                        rhfo |
                  At least 1 |   1.303559   .1769124     7.37   0.000     .9568167
>       1.6503
                             |
                        ghto |
                  At least 1 |  -.7309652   .1936978    -3.77   0.000    -1.110606
>    -.3513244
                             |
                    rhfo#ghto |
      At least 1#At least 1 |  -1.078837    .356254    -3.03   0.002    -1.777082
>    -.3805916
                             |
                     country |
                     Austria |   .5220959   .2161634     2.42   0.016     .0984235
>     .9457684
                     Germany |   1.835494   .2565373     7.15   0.000      1.33269
>     2.338298
                       Spain |   2.179997   .2613559     8.34   0.000     1.667749
>     2.692246
                       Italy |   1.686105   .2459653     6.86   0.000     1.204021
```

```
>     2.168188
                        France |   1.604541   .2179356     7.36   0.000     1.177395
>     2.031687
                       Denmark |  -.4317961    .273662    -1.58   0.115    -.9681637
>     .1045716
                   Switzerland |   .3512927    .254175     1.38   0.167    -.1468811
>     .8494666
                       Belgium |   .2947244   .2102657     1.40   0.161    -.1173887
>     .7068376
                 Czech Republic |   1.862402   .2784955     6.69   0.000      1.31656
>     2.408243
                       Slovenia |    1.95092   .3083791     6.33   0.000     1.346508
>     2.555332
----------------------------------------------------------------------------------------
  ___________
                        /phi1_1 |          1  (constrained)
                        /phi1_2 |    .880773   .0253766    34.71   0.000     .8310358
>     .9305103
                        /phi1_3 |     .66138   .0192976    34.27   0.000     .6235575
>     .6992025
                        /phi1_4 |   .3579472   .0165609    21.61   0.000     .3254886
>     .3904059
                        /phi1_5 |          0  (base outcome)
----------------------------------------------------------------------------------------
  ___________
                        /theta1 |   21.35913   6.552657     3.26   0.001     8.516161
>     34.20211
                        /theta2 |   20.23835   5.763686     3.51   0.000     8.941734
>     31.53497
                        /theta3 |   17.04334   4.356196     3.91   0.000     8.505354
>     25.58133
                        /theta4 |   10.10822   2.413491     4.19   0.000     5.377866
>     14.83858
                        /theta5 |          0  (base outcome)
----------------------------------------------------------------------------------------
  ___________
(sphus=Poor is the base outcome)
results_imp1_ste_15_4.xls
dir : seeout

Predictive margins                                      Number of obs = 18,545
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

----------------------------------------------------------------------------------------
  ____
                            |            Delta-method
                            |     Margin   std. err.      z    P>|z|     [95% conf. inter
> val]
----------------------------------------------------------------------------------------
  ____
      _predict#rhfo#ghto    |
            1#None#None     |   .0422315   .0024056    17.56   0.000     .0375166    .046
> 9464
      1#None#At least 1     |   .0523556   .0033282    15.73   0.000     .0458325    .058
> 8787
      1#At least 1#None     |   .0278092   .0023702    11.73   0.000     .0231637    .032
> 4548
1#At least 1#At least 1     |    .049078   .0046471    10.56   0.000     .0399698    .058
> 1862
            2#None#None     |    .124188   .0040614    30.58   0.000     .1162278    .132
> 1483
      2#None#At least 1     |   .1441812   .0060804    23.71   0.000     .1322639    .156
> 0985
      2#At least 1#None     |   .0920363   .0048934    18.81   0.000     .0824455    .101
> 6271
2#At least 1#At least 1     |   .1379026   .0080147    17.21   0.000     .1221941    .153
```

```
> 6112
            3#None#None |   .3879858   .0062522    62.06   0.000     .3757317    .400
> 2399
      3#None#At least 1 |   .4065464    .007923    51.31   0.000     .3910176    .422
> 0753
      3#At least 1#None |   .3464479   .0069858    49.59   0.000      .332756    .360
> 1398
3#At least 1#At least 1 |   .4012519   .0089073    45.05   0.000     .3837939    .418
> 7098
            4#None#None |   .3323593   .0058616    56.70   0.000     .3208707    .343
> 8478
      4#None#At least 1 |   .3085365   .0078324    39.39   0.000     .2931852    .323
> 8879
      4#At least 1#None |    .365724   .0074588    49.03   0.000     .3511051    .380
> 3429
4#At least 1#At least 1 |   .3161937   .0102062    30.98   0.000     .2961901    .336
> 1974
            5#None#None |   .1132354   .0042113    26.89   0.000     .1049814    .121
> 4894
      5#None#At least 1 |   .0883803   .0063323    13.96   0.000     .0759692    .100
> 7914
      5#At least 1#None |   .1679826   .0074528    22.54   0.000     .1533754    .182
> 5898
5#At least 1#At least 1 |   .0955738   .0090967    10.51   0.000     .0777446    .113
> 4029
                        |
------------------------+----------------------------------------------------------------
  ____
slog15_4.xls
dir : seeout
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.
note: 2.ghto omitted because of collinearity.
note: 0.rhfo#2.ghto identifies no observations in the sample.
note: 1.rhfo#2.ghto identifies no observations in the sample.
note: 2.rhfo#0.ghto identifies no observations in the sample.
note: 2.rhfo#1.ghto identifies no observations in the sample.
note: 2.rhfo#2.ghto omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -2.123e+08  (not concave)
Iteration 1:  Log pseudolikelihood =  -94922967  (not concave)
Iteration 2:  Log pseudolikelihood =  -58543734  (not concave)
Iteration 3:  Log pseudolikelihood =  -45853234  (not concave)
Iteration 4:  Log pseudolikelihood =  -42682207  (not concave)
Iteration 5:  Log pseudolikelihood =  -42317772  (not concave)
Iteration 6:  Log pseudolikelihood =  -42139943  (not concave)
Iteration 7:  Log pseudolikelihood =  -42027409  (not concave)
Iteration 8:  Log pseudolikelihood =  -41966530  (not concave)
Iteration 9:  Log pseudolikelihood =  -41939231
Iteration 10: Log pseudolikelihood =  -41920352  (not concave)
Iteration 11: Log pseudolikelihood =  -41838126  (not concave)
Iteration 12: Log pseudolikelihood =  -41824240
Iteration 13: Log pseudolikelihood =  -41812815
Iteration 14: Log pseudolikelihood =  -41811445
Iteration 15: Log pseudolikelihood =  -41811397
Iteration 16: Log pseudolikelihood =  -41811397

Stereotype logistic regression                          Number of obs = 16,181
                                                        Wald chi2(33) = 830.79
Log pseudolikelihood = -41811397                        Prob > chi2   = 0.0000

 ( 1)  [phi1_1]_cons = 1
------------------------------------------------------------------------------
  ____________
                                |               Robust
                          sphus | Coefficient  std. err.      z    P>|z|     [95% con
> f. interval]                   |
------------------------------------------------------------------------------
  ____________
                            age |   .3074247    .181892     1.69   0.091    -.0490772
>     .6639265                  |
                      c.age#c.age |  -.0016312   .0011755    -1.39   0.165     -.003935
```

```
>     .0006727
                                  |
                            yedu  |  -.0584023   .0519199    -1.12   0.261    -.1601634
>     .0433587
                                  |
                  c.yedu#c.yedu   |  -.0016207   .0024718    -0.66   0.512    -.0064653
>      .003224
                                  |
                           thinc  |  -.2213483   .1604577    -1.38   0.168    -.5358397
>     .0931431
                                  |
                 c.thinc#c.thinc  |  -.0778361   .0891909    -0.87   0.383    -.2526471
>     .0969749
                                  |
                          gender  |
                          Female  |   -.206216   .1370954    -1.50   0.133    -.4749181
>     .0624861
                          0.covid |          0  (omitted)
                                  |
                       household  |
        Couple, both responding   |   .1157886   .1481681     0.78   0.435    -.1746154
>     .4061927
Multiple, at least 1 responding   |    .093554   .2715963     0.34   0.731    -.4387649
>     .6258728
                                  |
                             cjs  |
                 Not applicable   |   3.024918   .7061962     4.28   0.000     1.640799
>     4.409037
                         Retired  |   1.312757   .5382786     2.44   0.015     .2577509
>     2.367764
                      Unemployed  |   .9640867   1.752274     0.55   0.582    -2.470308
>     4.398481
   Permanently sick or disabled   |   4.470294   .8578574     5.21   0.000     2.788924
>     6.151663
                       Homemaker  |    1.59884    .571506     2.80   0.005     .4787089
>     2.718971
                           Other  |   1.043749   .7254078     1.44   0.150    -.3780246
>     2.465522
                                  |
                            gali  |
                         Limited  |   4.787796   .2139706    22.38   0.000     4.368421
>     5.207171
                          lifesat |  -.3608914   .2084158    -1.73   0.083     -.769379
>     .0475961
                                  |
             c.lifesat#c.lifesat  |  -.0209828   .0148348    -1.41   0.157    -.0500586
>     .0080929
                                  |
                            naly  |
                       At least 1 |  -.3531956   .1664367    -2.12   0.034    -.6794055
>    -.0269857
                 Not applicable   |          0  (omitted)
                                  |
                            rhfo  |
                       At least 1 |   .8390433   .2319218     3.62   0.000     .3844849
>     1.293602
                 Not applicable   |   .1734152   .2456047     0.71   0.480    -.3079613
>     .6547916
                                  |
                            ghto  |
                       At least 1 |  -.7411539   .2339545    -3.17   0.002    -1.199696
>    -.2826115
                 Not applicable   |          0  (omitted)
                                  |
                       rhfo#ghto  |
            None#Not applicable   |          0  (empty)
          At least 1#At least 1   |   .7696974   .4239729     1.82   0.069    -.0612741
>     1.600669
      At least 1#Not applicable   |          0  (empty)
            Not applicable#None   |          0  (empty)
      Not applicable#At least 1   |          0  (empty)
  Not applicable#Not applicable   |          0  (omitted)
```

```
                         |
                 country |
                 Austria |   .6873048   .2257205     3.04   0.002     .2449008
>     1.129709           |
                 Germany |   1.620277   .2648329     6.12   0.000     1.101214
>      2.13934           |
                   Spain |    2.09674   .2669919     7.85   0.000     1.573445
>     2.620034           |
                   Italy |   1.620135   .2495526     6.49   0.000     1.131021
>     2.109249           |
                  France |   1.110426    .222524     4.99   0.000     .6742868
>     1.546565           |
                 Denmark |  -.7659217    .268393    -2.85   0.004    -1.291962
>    -.2398812           |
             Switzerland |  -.0190542   .2367687    -0.08   0.936    -.4831123
>     .4450038           |
                 Belgium |   .0284621   .2103363     0.14   0.892    -.3837894
>     .4407135           |
           Czech Republic |   .2801487   .2402226     1.17   0.244    -.1906789
>     .7509763           |
                Slovenia |   1.642754   .2829158     5.81   0.000      1.08825
>     2.197259           |
-------------------------+----------------------------------------------------------
                         |
            /phi1_1      |          1  (constrained)
            /phi1_2      |   .916075   .0321428    28.50   0.000     .8530763
>     .9790737           |
            /phi1_3      |   .6693081   .0233631    28.65   0.000     .6235172
>      .715099           |
            /phi1_4      |   .3437594   .0198103    17.35   0.000      .304932
>     .3825868           |
            /phi1_5      |          0  (base outcome)
-------------------------+----------------------------------------------------------
                         |
            /theta1      |   13.05881   7.056386     1.85   0.064    -.7714528
>     26.88907           |
            /theta2      |   13.23139    6.49759     2.04   0.042     .4963448
>     25.96643           |
            /theta3      |   11.45767   4.759877     2.41   0.016     2.128485
>     20.78686           |
            /theta4      |   6.701615   2.499225     2.68   0.007     1.803224
>     11.60001           |
            /theta5      |          0  (base outcome)
-------------------------+----------------------------------------------------------
(sphus=Poor is the base outcome)
results_imp1_ste_17_4.xls
dir : seeout

Predictive margins                                      Number of obs = 16,181
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

-------------------------+----------------------------------------------------------
                         |            Delta-method
                         |     Margin   std. err.      z    P>|z|     [95% con
> f. interval]            |
-------------------------+----------------------------------------------------------
                         |
  _predict#rhfo#ghto     |
          1#None#None    |   .040184   .0026065    15.42   0.000     .0350753
>     .0452927           |
     1#None#At least 1   |   .0502418   .0040009    12.56   0.000     .0424002
>     .0580833           |
 1#None#Not applicable   |          .  (not estimable)
     1#At least 1#None   |   .0305722   .0028678    10.66   0.000     .0249513
```

```
>      .036193
        1#At least 1#At least 1 |    .030277   .0037022     8.18   0.000     .0230207
>     .0375332
    1#At least 1#Not applicable |          .  (not estimable)
          1#Not applicable#None |          .  (not estimable)
    1#Not applicable#At least 1 |          .  (not estimable)
1#Not applicable#Not applicable |   .0380462   .0029403    12.94   0.000     .0322834
>     .0438091
                     2#None#None |   .1190707   .0050296    23.67   0.000     .1092129
>     .1289286
              2#None#At least 1 |   .1420035   .0081707    17.38   0.000     .1259892
>     .1580177
          2#None#Not applicable |          .  (not estimable)
              2#At least 1#None |   .0956124   .0064718    14.77   0.000     .0829279
>     .1082969
        2#At least 1#At least 1 |    .094864   .0084438    11.23   0.000     .0783145
>     .1114136
    2#At least 1#Not applicable |          .  (not estimable)
          2#Not applicable#None |          .  (not estimable)
    2#Not applicable#At least 1 |          .  (not estimable)
2#Not applicable#Not applicable |   .1139957    .006319    18.04   0.000     .1016107
>     .1263807
                     3#None#None |   .3824718   .0073646    51.93   0.000     .3680374
>     .3969062
              3#None#At least 1 |   .4041721   .0094777    42.64   0.000     .3855962
>      .422748
          3#None#Not applicable |          .  (not estimable)
              3#At least 1#None |   .3528548   .0094863    37.20   0.000      .334262
>     .3714476
        3#At least 1#At least 1 |   .3517664   .0133625    26.32   0.000     .3255764
>     .3779565
    3#At least 1#Not applicable |          .  (not estimable)
          3#Not applicable#None |          .  (not estimable)
    3#Not applicable#At least 1 |          .  (not estimable)
3#Not applicable#Not applicable |   .3767552   .0078762    47.83   0.000     .3613181
>     .3921923
                     4#None#None |   .3332366   .0071632    46.52   0.000      .319197
>     .3472762
              4#None#At least 1 |   .3066672   .0103559    29.61   0.000     .2863701
>     .3269644
          4#None#Not applicable |          .  (not estimable)
              4#At least 1#None |   .3583832   .0086765    41.31   0.000     .3413776
>     .3753887
        4#At least 1#At least 1 |   .3591335   .0105059    34.18   0.000     .3385424
>     .3797246
    4#At least 1#Not applicable |          .  (not estimable)
          4#Not applicable#None |          .  (not estimable)
    4#Not applicable#At least 1 |          .  (not estimable)
4#Not applicable#Not applicable |   .3389003   .0077802    43.56   0.000     .3236515
>     .3541491
                     5#None#None |   .1250368   .0058631    21.33   0.000     .1135454
>     .1365283
              5#None#At least 1 |   .0969154   .0084736    11.44   0.000     .0803075
>     .1135233
          5#None#Not applicable |          .  (not estimable)
              5#At least 1#None |   .1625775   .0106233    15.30   0.000     .1417562
>     .1833987
        5#At least 1#At least 1 |    .163959    .015561    10.54   0.000     .1334601
>      .194458
    5#At least 1#Not applicable |          .  (not estimable)
          5#Not applicable#None |          .  (not estimable)
    5#Not applicable#At least 1 |          .  (not estimable)
5#Not applicable#Not applicable |   .1323026   .0081711    16.19   0.000     .1162874
>     .1483177
------------------------------------------------------------------------------------------
```

slog17_4.xls
dir : seeout
note: **0.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -2.407e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.389e+08  (not concave)
Iteration 2:  Log pseudolikelihood =  -75267100  (not concave)
Iteration 3:  Log pseudolikelihood =  -42773934  (not concave)
Iteration 4:  Log pseudolikelihood =  -37366799  (not concave)
Iteration 5:  Log pseudolikelihood =  -31855044  (not concave)
Iteration 6:  Log pseudolikelihood =  -30246491  (not concave)
Iteration 7:  Log pseudolikelihood =  -29892418  (not concave)
Iteration 8:  Log pseudolikelihood =  -29699562  (not concave)
Iteration 9:  Log pseudolikelihood =  -29587695  (not concave)
Iteration 10: Log pseudolikelihood =  -29462779  (not concave)
Iteration 11: Log pseudolikelihood =  -29361629  (not concave)
Iteration 12: Log pseudolikelihood =  -29305878  (not concave)
Iteration 13: Log pseudolikelihood =  -29242681  (not concave)
Iteration 14: Log pseudolikelihood =  -29209163  (not concave)
Iteration 15: Log pseudolikelihood =  -29165671  (not concave)
Iteration 16: Log pseudolikelihood =  -29137409  (not concave)
Iteration 17: Log pseudolikelihood =  -29109230  (not concave)
Iteration 18: Log pseudolikelihood =  -28955918  (not concave)
Iteration 19: Log pseudolikelihood =  -28830741  (not concave)
Iteration 20: Log pseudolikelihood =  -28769110  (not concave)
Iteration 21: Log pseudolikelihood =  -28739619  (not concave)
Iteration 22: Log pseudolikelihood =  -28724708  (not concave)
Iteration 23: Log pseudolikelihood =  -28713230  (not concave)
Iteration 24: Log pseudolikelihood =  -28703542  (not concave)
Iteration 25: Log pseudolikelihood =  -28694921  (not concave)
Iteration 26: Log pseudolikelihood =  -28687181  (not concave)
Iteration 27: Log pseudolikelihood =  -28679829  (not concave)
Iteration 28: Log pseudolikelihood =  -28672611  (not concave)
Iteration 29: Log pseudolikelihood =  -28665580  (not concave)
Iteration 30: Log pseudolikelihood =  -28658724  (not concave)
Iteration 31: Log pseudolikelihood =  -28652004  (not concave)
Iteration 32: Log pseudolikelihood =  -28645404  (not concave)
Iteration 33: Log pseudolikelihood =  -28638915  (not concave)
Iteration 34: Log pseudolikelihood =  -28632522  (not concave)
Iteration 35: Log pseudolikelihood =  -28626209  (not concave)
Iteration 36: Log pseudolikelihood =  -28619962  (not concave)
Iteration 37: Log pseudolikelihood =  -28613778  (not concave)
Iteration 38: Log pseudolikelihood =  -28607641  (not concave)
Iteration 39: Log pseudolikelihood =  -28601552  (not concave)
Iteration 40: Log pseudolikelihood =  -28595497  (not concave)
Iteration 41: Log pseudolikelihood =  -28589480  (not concave)
Iteration 42: Log pseudolikelihood =  -28583488  (not concave)
Iteration 43: Log pseudolikelihood =  -28577527  (not concave)
Iteration 44: Log pseudolikelihood =  -28571585  (not concave)
Iteration 45: Log pseudolikelihood =  -28565668  (not concave)
Iteration 46: Log pseudolikelihood =  -28559765  (not concave)
Iteration 47: Log pseudolikelihood =  -28553883  (not concave)
Iteration 48: Log pseudolikelihood =  -28548011  (not concave)
Iteration 49: Log pseudolikelihood =  -28542157  (not concave)
Iteration 50: Log pseudolikelihood =  -28536310  (not concave)
Iteration 51: Log pseudolikelihood =  -28530477  (not concave)
Iteration 52: Log pseudolikelihood =  -28524650  (not concave)
Iteration 53: Log pseudolikelihood =  -28518835  (not concave)
Iteration 54: Log pseudolikelihood =  -28513023  (not concave)
Iteration 55: Log pseudolikelihood =  -28507222  (not concave)
Iteration 56: Log pseudolikelihood =  -28501423  (not concave)
Iteration 57: Log pseudolikelihood =  -28495633  (not concave)
Iteration 58: Log pseudolikelihood =  -28489844  (not concave)
Iteration 59: Log pseudolikelihood =  -28484063  (not concave)
Iteration 60: Log pseudolikelihood =  -28478282  (not concave)
Iteration 61: Log pseudolikelihood =  -28472509  (not concave)
Iteration 62: Log pseudolikelihood =  -28466735  (not concave)
Iteration 63: Log pseudolikelihood =  -28460969  (not concave)
Iteration 64: Log pseudolikelihood =  -28455201  (not concave)
Iteration 65: Log pseudolikelihood =  -28449442  (not concave)
Iteration 66: Log pseudolikelihood =  -28443680  (not concave)
Iteration 67: Log pseudolikelihood =  -28437927  (not concave)
Iteration 68: Log pseudolikelihood =  -28432172  (not concave)
Iteration 69: Log pseudolikelihood =  -28426426  (not concave)
Iteration 70: Log pseudolikelihood =  -28420678  (not concave)
Iteration 71: Log pseudolikelihood =  -28414939  (not concave)
```

```
Iteration 72: Log pseudolikelihood =  -28409200  (not concave)
Iteration 73: Log pseudolikelihood =  -28403469  (not concave)
Iteration 74: Log pseudolikelihood =  -28397739  (not concave)
Iteration 75: Log pseudolikelihood =  -28392018  (not concave)
Iteration 76: Log pseudolikelihood =  -28386299  (not concave)
Iteration 77: Log pseudolikelihood =  -28380589  (not concave)
Iteration 78: Log pseudolikelihood =  -28374882  (not concave)
Iteration 79: Log pseudolikelihood =  -28369186  (not concave)
Iteration 80: Log pseudolikelihood =  -28363493  (not concave)
Iteration 81: Log pseudolikelihood =  -28357813  (not concave)
Iteration 82: Log pseudolikelihood =  -28352137  (not concave)
Iteration 83: Log pseudolikelihood =  -28346474  (not concave)
Iteration 84: Log pseudolikelihood =  -28340816  (not concave)
Iteration 85: Log pseudolikelihood =  -28335173  (not concave)
Iteration 86: Log pseudolikelihood =  -28329537  (not concave)
Iteration 87: Log pseudolikelihood =  -28323917  (not concave)
Iteration 88: Log pseudolikelihood =  -28318305  (not concave)
Iteration 89: Log pseudolikelihood =  -28312709  (not concave)
Iteration 90: Log pseudolikelihood =  -28307124  (not concave)
Iteration 91: Log pseudolikelihood =  -28301557  (not concave)
Iteration 92: Log pseudolikelihood =  -28296001  (not concave)
Iteration 93: Log pseudolikelihood =  -28290465  (not concave)
Iteration 94: Log pseudolikelihood =  -28284942  (not concave)
Iteration 95: Log pseudolikelihood =  -28279440  (not concave)
Iteration 96: Log pseudolikelihood =  -28273952  (not concave)
Iteration 97: Log pseudolikelihood =  -28268487  (not concave)
Iteration 98: Log pseudolikelihood =  -28263039  (not concave)
Iteration 99: Log pseudolikelihood =  -28257614  (not concave)
Iteration 100: Log pseudolikelihood =  -28252207  (not concave)
Iteration 101: Log pseudolikelihood =  -28246826  (not concave)
Iteration 102: Log pseudolikelihood =  -28241464  (not concave)
Iteration 103: Log pseudolikelihood =  -28236130  (not concave)
Iteration 104: Log pseudolikelihood =  -28230816  (not concave)
Iteration 105: Log pseudolikelihood =  -28225531  (not concave)
Iteration 106: Log pseudolikelihood =  -28220269  (not concave)
Iteration 107: Log pseudolikelihood =  -28215038  (not concave)
Iteration 108: Log pseudolikelihood =  -28209831  (not concave)
Iteration 109: Log pseudolikelihood =  -28204655  (not concave)
Iteration 110: Log pseudolikelihood =  -28199506  (not concave)
Iteration 111: Log pseudolikelihood =  -28194390  (not concave)
Iteration 112: Log pseudolikelihood =  -28189301  (not concave)
Iteration 113: Log pseudolikelihood =  -28184248  (not concave)
Iteration 114: Log pseudolikelihood =  -28179223  (not concave)
Iteration 115: Log pseudolikelihood =  -28174235  (not concave)
Iteration 116: Log pseudolikelihood =  -28169278  (not concave)
Iteration 117: Log pseudolikelihood =  -28164358  (not concave)
Iteration 118: Log pseudolikelihood =  -28159470  (not concave)
Iteration 119: Log pseudolikelihood =  -28154622  (not concave)
Iteration 120: Log pseudolikelihood =  -28149807  (not concave)
Iteration 121: Log pseudolikelihood =  -28145032
Iteration 122: Log pseudolikelihood =  -27956546  (not concave)
Iteration 123: Log pseudolikelihood =  -27933284
Iteration 124: Log pseudolikelihood =  -27901477
Iteration 125: Log pseudolikelihood =  -27872404  (not concave)
Iteration 126: Log pseudolikelihood =  -27858991
Iteration 127: Log pseudolikelihood =  -27839461
Iteration 128: Log pseudolikelihood =  -27839366  (not concave)
Iteration 129: Log pseudolikelihood =  -27833785
Iteration 130: Log pseudolikelihood =  -27831175
Iteration 131: Log pseudolikelihood =  -27831056
Iteration 132: Log pseudolikelihood =  -27830311
Iteration 133: Log pseudolikelihood =  -27830290
Iteration 134: Log pseudolikelihood =  -27830290

Stereotype logistic regression                        Number of obs = 10,710
                                                      Wald chi2(33) = 720.44
Log pseudolikelihood = -27830290                      Prob > chi2   = 0.0000
```

(1) **[phi1_1]_cons = 1**

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.6042228	.2580341	2.34	0.019	.0984853	1.10996
c.age#c.age	-.0035689	.0016149	-2.21	0.027	-.006734	-.0004038
yedu	-.0270896	.0636792	-0.43	0.671	-.1518985	.0977192
c.yedu#c.yedu	-.0032098	.0029319	-1.09	0.274	-.0089563	.0025367
thinc	-.1639399	.1853363	-0.88	0.376	-.5271924	.1993125
c.thinc#c.thinc	.0075906	.0891563	0.09	0.932	-.1671525	.1823337
gender						
Female	.0064674	.1806182	0.04	0.971	-.3475378	.3604726
0.covid	0	(omitted)				
household						
Couple, both responding	.5141661	.2062223	2.49	0.013	.1099778	.9183543
Multiple, at least 1 responding	.5021928	.3351912	1.50	0.134	-.1547698	1.159156
cjs						
Not applicable	3.74138	1.263556	2.96	0.003	1.264856	6.217903
Retired	3.783053	1.058462	3.57	0.000	1.708505	5.8576
Unemployed	2.776052	2.536829	1.09	0.274	-2.196041	7.748144
Permanently sick or disabled	5.642394	1.336596	4.22	0.000	3.022714	8.262074
Homemaker	3.370497	1.087878	3.10	0.002	1.238296	5.502698
Other	3.623106	1.179401	3.07	0.002	1.311523	5.93469
gali						
Limited	5.329378	.2993792	17.80	0.000	4.742605	5.91615
lifesat	-1.584582	.3286304	-4.82	0.000	-2.228686	-.9404784
c.lifesat#c.lifesat	.0535431	.022751	2.35	0.019	.008952	.0981342
naly						
At least 1	-1.039182	.265969	-3.91	0.000	-1.560472	-.5178926
Not applicable	0	(omitted)				
rhfo						
At least 1	.8421124	.2311818	3.64	0.000	.3890043	1.29522
ghto						
At least 1	-.5677653	.2410207	-2.36	0.018	-1.040157	-.0953734

```
                                 |
                      rhfo#ghto  |
         At least 1#At least 1   |  -.2092123   .4672146    -0.45   0.654    -1.124936
>     .7065115
                                 |
                       country   |
                       Austria   |   .447301   .3021511     1.48   0.139    -.1449042
>     1.039506
                       Germany   |  1.198908   .3408866     3.52   0.000     .5307828
>     1.867034
                   Netherlands   | -.1930181   .3240486    -0.60   0.551    -.8281416
>     .4421054
                         Spain   |  2.166777   .3658821     5.92   0.000     1.449661
>     2.883893
                         Italy   |  1.329021   .3630353     3.66   0.000     .6174852
>     2.040557
                        France   |  .9870299   .2875559     3.43   0.001     .4234307
>     1.550629
                       Denmark   | -.4007072   .3492788    -1.15   0.251    -1.085281
>     .2838668
                   Switzerland   | -.2580571   .3363431    -0.77   0.443    -.9172775
>     .4011633
                       Belgium   |  .1209022   .2980841     0.41   0.685    -.4633318
>     .7051363
                 Czech Republic  | -.2151099     .34363    -0.63   0.531    -.8886123
>     .4583925
                       Slovenia  |  1.206378    .377758     3.19   0.001     .4659858
>      1.94677
-------------------------------------------------------------------------------------
  ____________
                       /phi1_1   |         1  (constrained)
                       /phi1_2   |  .8360516   .0392788    21.29   0.000     .7590664
>     .9130367
                       /phi1_3   |  .6390064    .029283    21.82   0.000     .5816128
>     .6963999
                       /phi1_4   |   .329027   .0232068    14.18   0.000     .2835425
>     .3745115
                       /phi1_5   |         0  (base outcome)
-------------------------------------------------------------------------------------
  ____________
                       /theta1   |   21.8206   10.28057     2.12   0.034      1.67105
>     41.97015
                       /theta2   |  19.93352   8.753742     2.28   0.023       2.7765
>     37.09054
                       /theta3   |  16.96167    6.65944     2.55   0.011     3.909411
>     30.01394
                       /theta4   |  9.671772   3.513104     2.75   0.006     2.786214
>     16.55733
                       /theta5   |         0  (base outcome)
-------------------------------------------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_20_4.xls
dir : seeout

Predictive margins                                      Number of obs = 10,710
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))
```

	Margin	Delta-method std. err.	z	P>|z|	[95% conf. interval]	
_predict#rhfo#ghto						
1#None#None	.036584	.0026738	13.68	0.000	.0313435	.0418244
1#None#At least 1	.0439093	.0038592	11.38	0.000	.0363454	.0514732
1#At least 1#None	.0274836	.0028037	9.80	0.000	.0219884	.0329788
1#At least 1#At least 1	.0358072	.0049404	7.25	0.000	.0261241	.0454903
2#None#None	.1299455	.0058394	22.25	0.000	.1185005	.1413906
2#None#At least 1	.1453706	.0078937	18.42	0.000	.1298992	.160842
2#At least 1#None	.1084517	.0064808	16.73	0.000	.0957496	.1211537
2#At least 1#At least 1	.1282208	.0107408	11.94	0.000	.1071693	.1492724
3#None#None	.3885574	.0081527	47.66	0.000	.3725783	.4045364
3#None#At least 1	.4046108	.0104122	38.86	0.000	.3842032	.4250184
3#At least 1#None	.3609708	.0095853	37.66	0.000	.342184	.3797576
3#At least 1#At least 1	.3865749	.0137137	28.19	0.000	.3596965	.4134532
4#None#None	.3409839	.0078577	43.40	0.000	.3255831	.3563846
4#None#At least 1	.3199757	.0107828	29.67	0.000	.2988419	.3411095
4#At least 1#None	.3682602	.0099025	37.19	0.000	.3488516	.3876688
4#At least 1#At least 1	.3432711	.0143674	23.89	0.000	.3151115	.3714307
5#None#None	.1039293	.0053525	19.42	0.000	.0934385	.1144201
5#None#At least 1	.0861335	.0074008	11.64	0.000	.0716283	.1006388
5#At least 1#None	.1348337	.0085149	15.83	0.000	.1181448	.1515227
5#At least 1#At least 1	.1061259	.0134113	7.91	0.000	.0798402	.1324117

slog20_4.xls
dir : seeout
note: **1.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -2.100e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.727e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.154e+08  (not concave)
Iteration 3:  Log pseudolikelihood =  -77910407  (not concave)
Iteration 4:  Log pseudolikelihood =  -58604075  (not concave)
Iteration 5:  Log pseudolikelihood =  -45862984  (not concave)
Iteration 6:  Log pseudolikelihood =  -43397708  (not concave)
Iteration 7:  Log pseudolikelihood =  -35053123  (not concave)
Iteration 8:  Log pseudolikelihood =  -33096876  (not concave)
Iteration 9:  Log pseudolikelihood =  -31975710  (not concave)
Iteration 10: Log pseudolikelihood =  -31172966  (not concave)
Iteration 11: Log pseudolikelihood =  -30683010  (not concave)
Iteration 12: Log pseudolikelihood =  -30405255  (not concave)
Iteration 13: Log pseudolikelihood =  -30181889  (not concave)
Iteration 14: Log pseudolikelihood =  -30016035  (not concave)
Iteration 15: Log pseudolikelihood =  -29893773  (not concave)
Iteration 16: Log pseudolikelihood =  -29811433  (not concave)
```

```
Iteration 17: Log pseudolikelihood =  -29733489  (not concave)
Iteration 18: Log pseudolikelihood =  -29427441  (not concave)
Iteration 19: Log pseudolikelihood =  -29331112  (not concave)
Iteration 20: Log pseudolikelihood =  -29286526  (not concave)
Iteration 21: Log pseudolikelihood =  -29239969  (not concave)
Iteration 22: Log pseudolikelihood =  -29199187  (not concave)
Iteration 23: Log pseudolikelihood =  -29161374  (not concave)
Iteration 24: Log pseudolikelihood =  -29122387  (not concave)
Iteration 25: Log pseudolikelihood =  -29083577  (not concave)
Iteration 26: Log pseudolikelihood =  -29045716  (not concave)
Iteration 27: Log pseudolikelihood =  -29008045  (not concave)
Iteration 28: Log pseudolikelihood =  -28970540  (not concave)
Iteration 29: Log pseudolikelihood =  -28933206  (not concave)
Iteration 30: Log pseudolikelihood =  -28896298  (not concave)
Iteration 31: Log pseudolikelihood =  -28859675  (not concave)
Iteration 32: Log pseudolikelihood =  -28823541  (not concave)
Iteration 33: Log pseudolikelihood =  -28787826  (not concave)
Iteration 34: Log pseudolikelihood =  -28752766  (not concave)
Iteration 35: Log pseudolikelihood =  -28718288  (not concave)
Iteration 36: Log pseudolikelihood =  -28684621  (not concave)
Iteration 37: Log pseudolikelihood =  -28651702  (not concave)
Iteration 38: Log pseudolikelihood =  -28619749  (not concave)
Iteration 39: Log pseudolikelihood =  -28588698  (not concave)
Iteration 40: Log pseudolikelihood =  -28558747  (not concave)
Iteration 41: Log pseudolikelihood =  -28529825  (not concave)
Iteration 42: Log pseudolikelihood =  -28502103  (not concave)
Iteration 43: Log pseudolikelihood =  -28475504
Iteration 44: Log pseudolikelihood =  -28427986  (not concave)
Iteration 45: Log pseudolikelihood =  -28333520
Iteration 46: Log pseudolikelihood =  -28245359  (not concave)
Iteration 47: Log pseudolikelihood =  -28211728
Iteration 48: Log pseudolikelihood =  -28152851  (not concave)
Iteration 49: Log pseudolikelihood =  -28140213
Iteration 50: Log pseudolikelihood =  -28096273  (not concave)
Iteration 51: Log pseudolikelihood =  -28091672
Iteration 52: Log pseudolikelihood =  -28087218  (not concave)
Iteration 53: Log pseudolikelihood =  -28075407
Iteration 54: Log pseudolikelihood =  -28073549
Iteration 55: Log pseudolikelihood =  -28073454
Iteration 56: Log pseudolikelihood =  -28073453
Iteration 57: Log pseudolikelihood =  -28073453

Stereotype logistic regression                          Number of obs = 11,204
                                                        Wald chi2(32) = 580.92
Log pseudolikelihood = -28073453                        Prob > chi2   = 0.0000

 ( 1)  [phi1_1]_cons = 1
------------------------------------------------------------------------------
             |               Robust
       sphus | Coefficient  std. err.      z    P>|z|     [95% con
> f. interval]
-------------+----------------------------------------------------------------
         age |   .9113787   .3102025     2.94   0.003     .3033928
>     1.519364
             |
 c.age#c.age |  -.0052964   .0019173    -2.76   0.006    -.0090542
>    -.0015386
             |
        yedu |   .0324176   .0661805     0.49   0.624    -.0972939
>      .162129
             |
 c.yedu#c.yedu |  -.0048227   .0029295    -1.65   0.100    -.0105644
>      .000919
             |
       thinc |  -.2403916    .195101    -1.23   0.218    -.6227825
>     .1419992
             |
 c.thinc#c.thinc |   .0080302   .0927774     0.09   0.931    -.1738103
>     .1898706
             |
```

```
                        gender |
                        Female |   .0953833    .182335     0.52   0.601    -.2619866
>     .4527533
                        1.covid |          0  (omitted)
                                |
                     household |
        Couple, both responding |   .6667631   .2078674     3.21   0.001     .2593505
>     1.074176
Multiple, at least 1 responding |    .323121   .4196699     0.77   0.441    -.4994169
>     1.145659
                                |
                           cjs |
                Not applicable |   -.302334    1.98332    -0.15   0.879    -4.189571
>     3.584903
                       Retired |  -.0404526   1.853557    -0.02   0.983    -3.673357
>     3.592452
   Permanently sick or disabled |   1.002965   2.102798     0.48   0.633    -3.118444
>     5.124375
                     Homemaker |  -.4547212   1.875574    -0.24   0.808     -4.13078
>     3.221337
                         Other |   .4923303   1.966451     0.25   0.802    -3.361844
>     4.346504
                                |
                          gali |
                       Limited |   5.717941   .3100638    18.44   0.000     5.110227
>     6.325655
                        lifesat |  -1.050432   .3056222    -3.44   0.001    -1.649441
>    -.4514238
                                |
             c.lifesat#c.lifesat |   .0059944   .0214193     0.28   0.780    -.0359866
>     .0479755
                                |
                          naly |
                     At least 1 |  -.9325415   .2308275    -4.04   0.000    -1.384955
>    -.4801278
                Not applicable |          0  (omitted)
                                |
                          rhfo |
                     At least 1 |    .760777    .222192     3.42   0.001     .3252888
>     1.196265
                                |
                          ghto |
                     At least 1 |  -.6437392   .2833046    -2.27   0.023    -1.199006
>    -.0884724
                                |
                      rhfo#ghto |
          At least 1#At least 1 |   .1635942     .50662     0.32   0.747    -.8293628
>     1.156551
                                |
                        country |
                        Austria |   .9696282   .3095917     3.13   0.002     .3628396
>     1.576417
                        Germany |   1.706057   .3671544     4.65   0.000     .9864474
>     2.425666
                    Netherlands |  -.1242691   .3446122    -0.36   0.718    -.7996966
>     .5511584
                          Spain |    2.29055   .3818893     6.00   0.000     1.542061
>      3.03904
                          Italy |    1.69037   .3579632     4.72   0.000     .9887754
>     2.391966
                         France |   1.107418    .309032     3.58   0.000     .5017261
>     1.713109
                        Denmark |  -.0401875   .3683501    -0.11   0.913    -.7621404
>     .6817653
                    Switzerland |   .1073201   .3764469     0.29   0.776    -.6305022
>     .8451425
                        Belgium |   .3158295   .3022631     1.04   0.296    -.2765952
>     .9082543
                  Czech Republic |   .1384658   .3779375     0.37   0.714    -.6022781
>     .8792096
                        Slovenia |   1.568619   .3615322     4.34   0.000     .8600294
>      2.27721
```

```
------------------------------------------------------------------------------
             /phi1_1 |          1  (constrained)
             /phi1_2 |   .8363109    .037478    22.31   0.000     .7628553
>     .9097665
             /phi1_3 |   .6327371    .029508    21.44   0.000     .5749026
>     .6905717
             /phi1_4 |   .3536328    .023918    14.79   0.000     .3067544
>     .4005112
             /phi1_5 |          0  (base outcome)
------------------------------------------------------------------------------
             /theta1 |   33.79511   12.66636     2.67   0.008     8.969496
>     58.62071
             /theta2 |   29.86759   10.51581     2.84   0.005     9.256979
>      50.4782
             /theta3 |     24.469   7.979718     3.07   0.002     8.829039
>     40.10896
             /theta4 |   14.52278    4.46304     3.25   0.001     5.775377
>     23.27017
             /theta5 |          0  (base outcome)
------------------------------------------------------------------------------
(sphus=Poor is the base outcome)
results_imp1_ste_22_4.xls
dir : seeout

Predictive margins                                      Number of obs = 11,204
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------------------------------------------------------
                 |            Delta-method
                 |     Margin   std. err.      z    P>|z|     [95% conf. inter
> val]
-----------------+------------------------------------------------------------
_predict#rhfo#ghto |
       1#None#None |   .0365861   .0026624    13.74   0.000     .0313679    .041
> 8042
   1#None#At least 1 |   .0450078   .0047488     9.48   0.000     .0357004    .054
> 3153
   1#At least 1#None |   .0282533   .0028619     9.87   0.000     .0226441    .033
> 8624
1#At least 1#At least 1 |   .0333172    .005002     6.66   0.000     .0235134     .04
> 3121
       2#None#None |   .1150155   .0048947    23.50   0.000      .105422    .124
> 6091
   2#None#At least 1 |   .1305187   .0078498    16.63   0.000     .1151333    .145
> 9041
   2#At least 1#None |   .0978334   .0056481    17.32   0.000     .0867633    .108
> 9035
2#At least 1#At least 1 |   .1085236   .0094969    11.43   0.000       .08991    .127
> 1372
       3#None#None |   .3904144   .0079319    49.22   0.000      .374868    .405
> 9607
   3#None#At least 1 |   .4081298   .0107575    37.94   0.000     .3870455    .429
> 2141
   3#At least 1#None |   .3663229   .0090822    40.33   0.000     .3485221    .384
> 1236
3#At least 1#At least 1 |   .3818908   .0137644    27.74   0.000     .3549131    .408
> 8684
       4#None#None |   .3425254   .0074166    46.18   0.000     .3279891    .357
> 0617
   4#None#At least 1 |   .3230729   .0108701    29.72   0.000      .301768    .344
> 3778
```

```
      4#At least 1#None |   .3616024   .0089835    40.25   0.000      .343995    .379
> 2098
4#At least 1#At least 1 |   .3500921   .0122612    28.55   0.000     .3260606    .374
> 1236
             5#None#None |   .1154586   .0054081    21.35   0.000     .1048589    .126
> 0584
      5#None#At least 1 |   .0932708   .0093081    10.02   0.000     .0750272    .111
> 5143
      5#At least 1#None |    .145988   .0084917    17.19   0.000     .1293445    .162
> 6315
5#At least 1#At least 1 |   .1261763   .0153662     8.21   0.000     .0960592    .156
> 2935
------------------------+----------------------------------------------------------------
  ____
slog22_4.xls
dir : seeout
      name:  The_value_of_a_helping_hand_sup
       log:  C:\Users\hansg\Documents\working\SHARE data\weighting\output_sup_4.smcl
  log type:  smcl
 closed on:   5 Apr 2026, 14:31:10
------------------------------------------------------------------------------------------
```